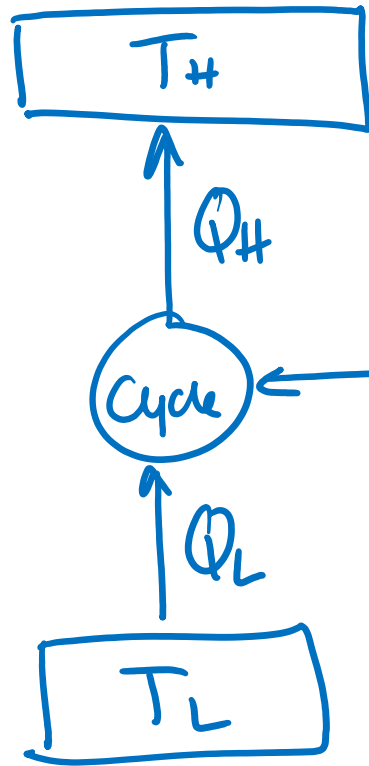




ME 300:
Engineering Thermodynamics

Heat Pumps and Refrigerators

Heat pumps



Goal: Heat T_H
: Tracking Q_H

Energy conservation:

$$Q_H = \bar{W}_{in} + Q_L$$

Coefficient of performance (COP)

$$\beta = \frac{\text{desired energy}}{\text{input energy}}$$

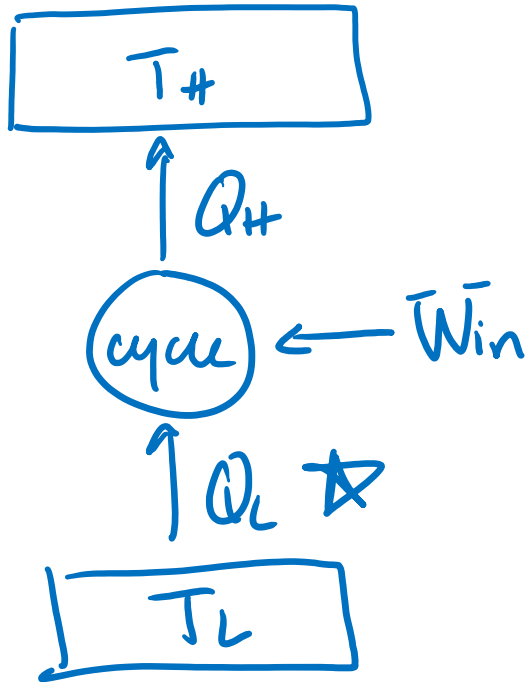
$$\beta_{\text{heat pump}} = \frac{Q_H}{\bar{W}_{in}}$$

$$\beta_{\text{Carnot, heat pump}} = \frac{T_H}{T_H - T_L}$$



What's the point of a refrigerator?

Refrigerators



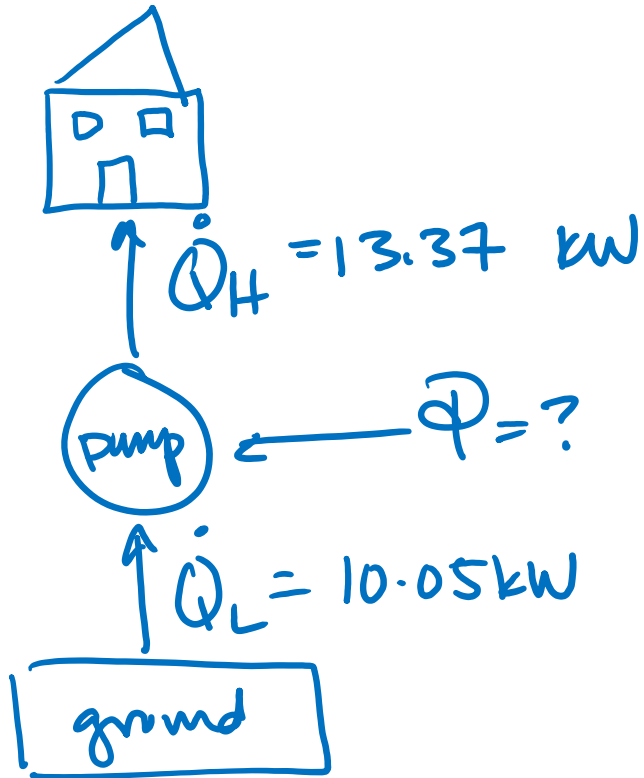
$$\beta = \frac{\text{desired energy}}{\text{input energy}}$$

$$\beta_{\text{frig}} = \frac{Q_L}{\bar{W}_{in}}$$

$$\beta_{\text{carnot, frig}} = \frac{T_L}{T_H - T_L}$$

Heat pump example

A residential heat pump uses the ground as an energy source. At a particular operating condition, the heat pump delivers energy at 13.37 kW to the interior while simultaneously removing energy from the ground at 10.05 kW. Determine the electrical power required to run the pump and the COP.



$$\dot{Q}_H = \dot{W} + \dot{Q}_L$$

$$\dot{W} = \dot{Q}_H - \dot{Q}_L = 13.37 - 10.05$$

$$\boxed{\dot{W} = 3.32 \text{ kW}}$$

$$\beta = \frac{\dot{Q}_H}{\dot{W}} = \frac{13.37}{3.32}$$

$$\rightarrow \boxed{\beta = 4.03}$$

Refrigerator example

A refrigerator operates with a COP of 4.2 and requires an electrical input of 700 W. Determine the rate of heat transfer from the cold space and the rate at which the device rejects heat.

$$\beta = 4.2$$
$$\dot{\Phi} = 700 \text{ W}$$

$$\dot{Q}_L = ?$$

$$\dot{Q}_H = ?$$

$$\beta = \frac{\dot{Q}_L}{\dot{\Phi}} \rightarrow \dot{Q}_L = \beta \dot{\Phi} = (4.2)(700)$$

$$\dot{Q}_L = 2.94 \text{ kW}$$

$$\dot{Q}_H = \dot{\Phi} + \dot{Q}_L$$

$$= 0.7 + 2.94 \text{ kW}$$

$$\dot{Q}_H = 3.64 \text{ kW}$$